

# Charles Langer

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## Education

### University of Utah, Salt Lake City

December 2026

Bachelor of Science in Electrical Engineering, GPA: 3.85

- Seven-Time Dean's List
- Co-Author, IEEE Manuscript Submission

### Courses Completed

- Classical Control Systems
- Linear & Active Circuit Design
- Cyber-Physical Systems
- Power System Analysis
- Introduction to Power
- Object-Oriented Programming
- Fundamental Electromagnetics
- PDEs for Engineers
- Power Economics/Operations

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## Technical Projects

### Traction Power Substation Design

Spring 2026

- Designed and reviewed single-line diagrams for AC utility interconnection and 750 V DC traction power systems, including Delta-Wye transformers and rectifier conversion
- Assisted in interpreting electrical drawings, and IEEE protective relay device numbers for breakers, fuses, instrument transformers, and multifunctional protective relays to ensure proper system coordination and code compliance
- Developed TPSS equipment layout and raceway/cable schedule for utility, transformer, rectifier, and feeder connections
- Sketched a Blue Light Station with an Emergency Trip System integrating PLCs to provide remote shutdown capability and coordinate protective devices with SCADA communications

### Modeling & Controlling of a Flexible System

Fall 2025

- Worked alongside a team to model a flexible crane-like system in MATLAB using Bode plots and dynamic analysis
- Programmed and tuned a PID controller using root locus, phase margin, steady-state error analysis to meet settling time and overshoot specifications
- Facilitated project coordination and consolidated modeling and control results into a formal technical report

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## Work Experience

### BAMBC Controls

May 2026 – August 2026

#### Electrical Engineering Intern

Burbank, CA

- Translated mechanical requirements into PLC-based control logic for a dual sump pump system.
- Engineered and built a low-cost control system prototype, including panel layout, wiring, PLC input and output connections, switches, and status indicators.
- Formulated lead/lag sequencing and interlocks using simulated field inputs to detect pump-flow failures.
- Devised and executed a testing procedure to verify PLC inputs, outputs, wiring, simulated field signals, interlocks and ladder-logic operation.

### Whole Foods

September 2022 – June 2023

#### Prepared Foods Team Member

Salt Lake City, UT

- Maintained quality control standards by monitoring inventory quality, supported end-of-day closeout procedures, and ensured adherence to established health, and safety standards.
- Demonstrated reliability and attention to detail by following structured procedures, executing assigned tasks accurately, and earning expanded responsibilities beyond a part-time scope.

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## Technical Skills

- Adept in MATLAB, Java, Python, and prompt engineering for AI-assisted research, analysis, and workflow efficiency.
- Familiar with NFPA 70-NEC, IEEE, NFPA-130 and technical design software programs such as AutoCAD & Revit
- Proficient with Microsoft Word, PowerPoint, and Excel

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## Affiliations

- Institute of Transportation Engineers Student Chapter Member
  - Vice-President of University of Utah Badminton Club
- July 2025 – Present  
May 2024 – Present